

USENGIMANA RICHARD

Data Analyst | Data Science Researcher | Applied Mathematics Graduate

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📍 Kigali, Rwanda

🌐 usengimanarichard531.github.io/newportfolio/

PROFILE

Data Science graduate student at AIMS Rwanda with a strong foundation in applied mathematics, spatial analysis, and statistical modelling. My current research applies multilevel regression, spatial autocorrelation, and satellite-derived environmental data to understand child malnutrition patterns across Rwanda's 30 districts work that sits at the intersection of data science and real-world public health impact. I am experienced in R, Python, and GIS tools, comfortable handling messy datasets, and committed to producing analysis that is reproducible, well-documented, and decision-relevant.

EXPERIENCE

Internship in Planning, Monitoring & Evaluation

Muhanga District

📅 2024 (1 month) 📍 Muhanga, Rwanda

- Gained hands-on experience in planning, monitoring, and evaluation activities.
- Supported data management processes related to population and administrative systems.
- Developed understanding of government reporting, local planning, and public service delivery.

Django Web Development Course

Solvit Africa

📅 2025 (3 months) 📍 Sonatube-Remera

- Completed intensive training in Django web development and Python-based applications.
- Built dynamic and data-driven web applications with database integration.
- Gained practical experience in APIs, user authentication, front-end integration, and scalable web solutions.

PROJECTS

Spatial Analysis of Child Malnutrition in Rwanda

AIMS Rwanda

- Conducting district-level analysis of child malnutrition using Rwanda Demographic and Health Survey data.
- Integrating environmental and socio-demographic factors such as rainfall, NDVI, altitude, maternal education, household wealth, water source, and sanitation.
- Applying descriptive analysis, spatial autocorrelation, regression models, and spatial regression techniques.
- Using R and Python for data cleaning, visualization, mapping, and statistical modeling.

SKILLS

Python R SQL PostgreSQL
Machine Learning Statistical Analysis

Data Cleaning Data Visualization
Spatial Analysis Regression Models
Research Writing

Microsoft Excel GitHub Linux LaTeX

TOOLS

Pandas NumPy Scikit-learn GeoPandas
R shiny for Darshboard development Excel
Power BI

RStudio ggplot2 sf spdep Django

EDUCATION

MSc. Mathematical Sciences, Specialization in Data Science

African Institute for Mathematical Sciences (AIMS) Rwanda

📅 Sept 2025 – June 2026 📍 Rwanda

- MasterCard Scholarship recipient.
- Specialized in Data Science, Machine Learning, Statistical Analysis, and AI.
- Research-focused training with practical applications in real-world data problems.

B.Sc. in Applied Mathematics

University of Rwanda, Kigali

📅 2022 – 2025 📍 Kigali, Rwanda

- Government of Rwanda Scholarship recipient.
- Focused on mathematical modeling, statistics, and computational methods.
- Final project on differential equations and mathematical analysis.

LANGUAGES

Kinyarwanda (Native) ●●●●●
• Native speaker with full professional proficiency.

English (Fluent) ●●●●●

Relational Database System for Insurance Policy and Claims Management

AIMS Rwanda – Database and Data Management

- Designed and implemented a fully normalized relational database system to manage insurance customers, policies, premiums, and claims processing.
- Modeled ten core entities including Policyholder, Policy, Claim, Adjuster, Payout, and Audit Log using Entity-Relationship (ER) diagram design.
- Transformed the ER model into a relational schema and implemented the full database using SQL, including table creation, foreign key constraints, and sample queries.

AI Nutrition Guidance Platform

SOLVIT Hackathon

- Working with a team to develop an AI-based platform that provides early nutrition guidance for Rwandans.
- Applying data-driven and AI approaches to support community health and nutrition awareness.
- Project selected among the top 15 teams to continue in the competition.

Intelligent Data Cleaning System

Personal Project

- Built an AI-powered system for automated data cleaning and preprocessing.
- Developed tools for data quality assessment, missing values, outliers, and inconsistent data formats.
- Created a user-friendly interface to help users prepare datasets for analysis and machine learning.

- Fluent in writing, reading, speaking, and listening.
- Able to communicate academic and professional work clearly.

REFERENCES

Dr. Lema Seknewna

@ Data Scientist & Professor at AIMS Rwanda

✉ lema.seknewna@aims.ac.rw

AIMS Rwanda

- Current supervisor for my Master's thesis.
- Can attest to my skills in data analysis and Python.

NYAGAHAKWA Venuste

@ Lecturer at University of Rwanda

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University of Rwanda

- Academic reference from undergraduate studies.
- Can attest to my mathematical and analytical abilities.